

JACOB D. CASTRO

Santa Maria, CA, USA | me@jacobdcastro.com | jacobdcastro.com | github.com/jacobdcastro | x.com/jacobdcastro

SKILLS

Programming	TypeScript, JavaScript, Python, Rust, SQL, Node.js
Frameworks	React, Next.js, GraphQL, TanStack Query, Zod, Tailwind, Vitest, Playwright, Storybook
AI & Pipelines	LLM workflows, agent tooling, prompt engineering, structured outputs, event-driven jobs
Infrastructure	Docker, PostgreSQL, GCP, AWS, Vercel, Railway, CI/CD, OpenTelemetry, Grafana, Sentry

EXPERIENCE

Senior Fullstack Engineer 01/2025 - Present
Phylax Systems Remote

- Led the fullstack TypeScript-based microservices platform: contract-first API design (130+ Zod-validated RESTful endpoints) with cross-service contract and version compatibility testing, multi-tenant PostgreSQL with row-level security, React UI, data pipelines, and CI/CD. Reviewed all teammate PRs and owned core features from ambiguous requirements through spec, implementation, and deployment.
- Architected an LLM agent service from the first commit, designing Python workflows, custom agent tools, generate-critique-refine quality loops, and deterministic verification of model output to power four AI features shipped to production.
- Designed and built the platform's event-driven data backbone: 40+ background workers, a GraphQL chain indexing layer, and a typed async job engine with retry policies and payload redaction, powering alerting, on-chain events, and AI workflows.
- Stack: TypeScript, Node.js, React, PostgreSQL, GraphQL, Zod, Docker

Freelancer 11/2018 - 01/2025
XIV Systems Remote

- Contributed full-stack improvement features for Sudoswap (sudoswap.xyz)
- Developed and deployed smart contracts, NFT minting app, and first e-commerce website for Highly Liquid (highlyliquidny.com)

Co-Founder and CEO 05/2024 - 01/2025
Monea Remote

- Built Monea's core product: a Rust-based CLI and engine for deploying rollup and Ethereum infrastructure.
- Led business development, including networking, partnerships, and fundraising strategy.
- Stack: Rust, Starlark, Docker

Lead Engineer 11/2022 - 07/2024
Brink Remote

- Developed [Brink V2 Intents Protocol](#), [CroutonSwap](#), [Brinkscan](#), [Nomial](#), and Brink L2 chain.
- Managed developer relations, documentation, and NPM releases; keynoted at ETHGlobal NYC and Devconnect Istanbul.
- Stack: JavaScript, TypeScript, Next.js, React, Redux, wagmi/ethers.js, Node.js, Solidity

Founding Engineer 04/2022 - 08/2022
Coinbooks (YC-backed) Remote

- Led development of a full-stack crypto invoicing app with native integration to core accounting product.
- Ran YC's 'write code and talk to users' playbook — met directly with paying customers, rapidly prototyped their requests, and folded their feedback into every release cycle.
- Stack: SPA React, TypeScript, Next.js, Ethers.js, Solidity, Express, MongoDB, AWS

Full-stack Developer 06/2021 - 04/2022
cocoNFT (Acquired) Remote

- Worked on an NFT minting and sales platform for creators, developing pixel-perfect UI based on designer's Figma files. (coconft.com)

PROJECTS

- [Preconf Devnet Dashboard](#) — ZuBerlin 2024 hackathon winner — real-time dashboard for Ethereum preconfirmation devnets with live block and validator metrics.
- [CHIP-8 CPU Emulator](#) — From-scratch CHIP-8 interpreter and emulator written in Rust with a minimal graphics frontend.
- [xiv — AI Agent Workflows](#) — Custom CLI orchestrating AI agent engineering workflows — automated implement, review, and ship loops with local review agents and overnight stacked-branch feature builds.
- [Personal Site](#) — Interactive portfolio built with TanStack Start, Three.js galaxy navigation, and a programmatic PDF resume generator.